

Can you leave a little trap?

Name: _____

Date: _____

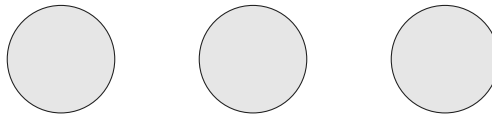
Two players alternate. Take **1 or 2** counters each turn. Never take more than remain. Taking the last counter wins.

1. Play first.

Try piles of 4 and 5 counters. Trade the first turn. Check every move.

2. A pile of three.

Choose first or second. If you choose second, show what you will do after **either** of the first player's moves.



If the first player takes 1, I will...

If the first player takes 2, I will...

3. Tell the whole story.

Does your plan work only when your partner helps, or against both choices? Demonstrate both branches with counters.

Build first. Explain by pointing, talking, or drawing.

A trap that grows

Name: _____

Date: _____

Two players alternate. Take **1 or 2** counters each turn. Never take more than remain. Taking the last counter wins.

4. Try six counters.

Arrange them as two groups of three. Can the second player answer every move and leave exactly three?



My two replies from six:

5. Choose first or second.

Try piles of 7, 8, and 9. Circle the piles you would like the **other player** to receive. Explain by playing.

7 8 9

What can I leave after the other player's turn and my reply?

Go further: Could the same plan work with three groups of three? Four groups? You can prove it by describing how one whole group disappears in each pair of turns.

Build first. Explain by pointing, talking, or drawing.